



第二章 矩阵代数

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§ 2.5.1 分块矩阵及其运算

一、矩阵的分块

对于行数和列数较高的矩阵 A ，为了简化运算，经常采用**分块法**，使大矩阵的运算化成小矩阵的运算。

具体做法：将矩阵 A 用若干条纵线和横线分成许多个**小矩阵**，每一个小矩阵称为 A 的**子块**，以子块为元素的形式上的矩阵称为**分块矩阵**。

例

$$A = \begin{pmatrix} a & 1 & 0 & 0 \\ 0 & a & 0 & 0 \\ 1 & 0 & b & 1 \\ 0 & 1 & 1 & b \end{pmatrix} = \begin{pmatrix} B_1 \\ B_2 \\ B_3 \end{pmatrix},$$

即

$$A = \begin{pmatrix} a & 1 & 0 & 0 \\ 0 & a & 0 & 0 \\ 1 & 0 & b & 1 \\ 0 & 1 & 1 & b \end{pmatrix} = \begin{pmatrix} B_1 \\ B_2 \\ B_3 \end{pmatrix}$$

$$A = \left(\begin{array}{cc|cc} a & 1 & 0 & 0 \\ 0 & a & 0 & 0 \\ 1 & 0 & b & 1 \\ 0 & 1 & 1 & b \end{array} \right) = \begin{pmatrix} C_1 & C_2 \\ C_3 & C_4 \end{pmatrix},$$

即

$$A = \begin{pmatrix} a & 1 & 0 & 0 \\ 0 & a & 0 & 0 \\ 1 & 0 & b & 1 \\ 0 & 1 & 1 & b \end{pmatrix} = \begin{pmatrix} C_1 & C_2 \\ C_3 & C_4 \end{pmatrix}$$

$$A = \begin{pmatrix} a & 1 & 0 & 0 \\ 0 & a & 0 & 0 \\ 1 & 0 & b & 1 \\ 0 & 1 & 1 & b \end{pmatrix} = \begin{pmatrix} A_1 & O \\ E & B \end{pmatrix}, \text{ 其中 } A_1 = \begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$$

$$A = \begin{pmatrix} a & 1 & 0 & 0 \\ 0 & a & 0 & 0 \\ 1 & 0 & b & 1 \\ 0 & 1 & 1 & b \end{pmatrix} = (A_1 \quad A_2 \quad A_3 \quad A_4), \text{ 其中 } A_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

二、分块矩阵的运算规则

(1) 设 A, B 为同型矩阵, 采用相同的分块法, 有

$$A = \begin{pmatrix} \boxed{A_{11}} & \cdots & \boxed{A_{1r}} \\ \vdots & & \vdots \\ A_{s1} & \cdots & A_{sr} \end{pmatrix}, \quad B = \begin{pmatrix} \boxed{B_{11}} & \cdots & \boxed{B_{1r}} \\ \vdots & & \vdots \\ B_{s1} & \cdots & B_{sr} \end{pmatrix}$$

其中 A_{ij} 与 B_{ij} 为同型矩阵, 那末

$$A + B = \begin{pmatrix} \boxed{A_{11} + B_{11}} & \cdots & \boxed{A_{1r} + B_{1r}} \\ \vdots & & \vdots \\ A_{s1} + B_{s1} & \cdots & A_{sr} + B_{sr} \end{pmatrix}.$$

即: $(A_{ij})_{sr} + (B_{ij})_{sr} = (A_{ij} + B_{ij})_{sr}$

(2) 设 $A = \begin{pmatrix} A_{11} & \cdots & A_{1r} \\ \vdots & & \vdots \\ A_{s1} & \cdots & A_{sr} \end{pmatrix}$, λ 为数, 那末

$$\lambda A = \begin{pmatrix} \lambda A_{11} & \cdots & \lambda A_{1r} \\ \vdots & & \vdots \\ \lambda A_{s1} & \cdots & \lambda A_{sr} \end{pmatrix}.$$

即: $\lambda(A_{ij})_{sr} = (\lambda A_{ij})_{sr}$

(3) 设 A 为 $m \times l$ 矩阵, B 为 $l \times n$ 矩阵, 分块成

$$A = \begin{pmatrix} A_{11} & \cdots & A_{1t} \\ \vdots & & \vdots \\ A_{s1} & \cdots & A_{st} \end{pmatrix}, \quad B = \begin{pmatrix} B_{11} & \cdots & B_{1r} \\ \vdots & & \vdots \\ B_{t1} & \cdots & B_{tr} \end{pmatrix},$$

其中 $A_{i1}, A_{i2}, \dots, A_{it}$ 的列数分别等于 $B_{1j}, B_{2j}, \dots, B_{tj}$ 的行数, 那末

$$AB = \begin{pmatrix} C_{11} & \cdots & C_{1r} \\ \vdots & & \vdots \\ C_{s1} & \cdots & C_{sr} \end{pmatrix}$$

其中 $C_{ij} = \sum_{k=1}^t A_{ik} B_{kj} \quad (i = 1, \dots, s; j = 1, \dots, r).$

$$\text{即 } (A_{ij})_{st} (B_{ij})_{tr} = (C_{ij})_{sr} = \left(\sum_{k=1}^t A_{ik} B_{kj} \right)_{sr}$$

(4) 设 $A = \begin{pmatrix} A_{11} & \cdots & A_{1r} \\ \vdots & & \vdots \\ A_{s1} & \cdots & A_{sr} \end{pmatrix}$, 则 $A^T = \begin{pmatrix} A_{11}^T & \cdots & A_{s1}^T \\ \vdots & & \vdots \\ A_{1r}^T & \cdots & A_{sr}^T \end{pmatrix}$.

(5) 设 A 为 n 阶矩阵, 若 A 的分块矩阵只有在主对角线上有非零子块, 其余子块都为零矩阵, 且非零子块都是方阵. 即

$$A = \begin{pmatrix} A_1 & & & \\ & A_2 & & \\ & & \ddots & \\ & & & A_s \end{pmatrix},$$

$$A = \begin{pmatrix} A_1 & & & \\ & A_2 & & \\ & & \ddots & \\ & & & A_s \end{pmatrix},$$

其中 A_i ($i = 1, 2, \dots, s$) 都是方阵, 那末称 A 为 **分块对角矩阵**. 记做 $\text{diag}(A_1, A_2, \dots, A_s)$. (准对角阵)

分块对角矩阵的行列式具有下述性质:

$$|A| = |A_1| |A_2| \cdots |A_s|.$$

(6) 设 $A = \begin{pmatrix} A_1 & & \mathbf{0} \\ & A_2 & \\ \mathbf{0} & & \ddots \\ & & & A_s \end{pmatrix},$

若 $|A_i| \neq 0 (i = 1, 2, \dots, s)$, 则 $|A| \neq 0$, 并有

$$A^{-1} = \begin{pmatrix} A_1^{-1} & & \mathbf{0} \\ & A_2^{-1} & \\ \mathbf{0} & & \ddots \\ & & & A_s^{-1} \end{pmatrix}.$$

即 $A^{-1} = \text{diag}(A_1^{-1}, A_2^{-1}, \dots, A_s^{-1})$

(7) 假设运算可行, 则

$$\begin{pmatrix} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & A_s \end{pmatrix} \begin{pmatrix} B_1 & 0 & \cdots & 0 \\ 0 & B_2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & B_s \end{pmatrix}$$
$$= \begin{pmatrix} A_1 B_1 & 0 & \cdots & 0 \\ 0 & A_2 B_2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & A_s B_s \end{pmatrix}.$$

即

$$\begin{aligned} AB &= \text{diag}(A_1, A_2, \cdots, A_s) \text{diag}(B_1, B_2, \cdots, B_s) \\ &= \text{diag}(A_1 B_1, A_2 B_2, \cdots, A_s B_s) \end{aligned}$$

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例1 设

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 1 & 0 \\ -1 & 2 & 0 & 1 \\ 1 & 0 & 4 & 1 \\ -1 & -1 & 2 & 0 \end{pmatrix},$$

求 AB .

解 把 A, B 分块成

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} E & O \\ A_1 & E \end{pmatrix},$$

$$B = \begin{pmatrix} 1 & 0 & 1 & 0 \\ -1 & 2 & 0 & 1 \\ 1 & 0 & 4 & 1 \\ -1 & -1 & 2 & 0 \end{pmatrix} = \begin{pmatrix} B_{11} & E \\ B_{21} & B_{22} \end{pmatrix}$$

则

$$AB = \begin{pmatrix} E & O \\ A_1 & E \end{pmatrix} \begin{pmatrix} B_{11} & E \\ B_{21} & B_{22} \end{pmatrix}$$

$$= \begin{pmatrix} B_{11} & E \\ A_1 B_{11} + B_{21} & A_1 + B_{22} \end{pmatrix}.$$

$$AB = \begin{pmatrix} B_{11} & E \\ A_1 B_{11} + B_{21} & A_1 + B_{22} \end{pmatrix}.$$

$$\begin{aligned} \text{又 } A_1 B_{11} + B_{21} &= \begin{pmatrix} -1 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix} \\ &= \begin{pmatrix} -3 & 4 \\ 0 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ -1 & 1 \end{pmatrix}, \end{aligned}$$

$$A_1 + B_{22} = \begin{pmatrix} -1 & 2 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 4 & 1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 3 & 1 \end{pmatrix},$$

于是

$$AB = \begin{pmatrix} B_{11} & E \\ A_1 B_{11} + B_{21} & A_1 + B_{22} \end{pmatrix}$$

$$= \left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ -1 & 4 & 0 & 1 \\ -2 & 4 & 3 & 3 \\ -1 & 1 & 3 & 1 \end{array} \right).$$

例2 设 $A = \begin{pmatrix} a & 1 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & b & 1 \\ 0 & 0 & 1 & b \end{pmatrix},$

$$B = \begin{pmatrix} a & 0 & 0 & 0 \\ 1 & a & 0 & 0 \\ 0 & 0 & b & 0 \\ 0 & 0 & 1 & b \end{pmatrix}$$

求 $A + B,$ $ABA.$

解 将 A, B 分块

$$A = \left(\begin{array}{cc|cc} a & 1 & 0 & 0 \\ 0 & a & 0 & 0 \\ \hline 0 & 0 & b & 1 \\ 0 & 0 & 1 & b \end{array} \right) = \begin{pmatrix} A_1 & O \\ O & A_2 \end{pmatrix}, \text{ 其中}$$

$$A_1 = \begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix},$$

$$A_2 = \begin{pmatrix} b & 1 \\ 1 & b \end{pmatrix};$$

$$B = \left(\begin{array}{cc|cc} a & 0 & 0 & 0 \\ 1 & a & 0 & 0 \\ \hline 0 & 0 & b & 0 \\ 0 & 0 & 1 & b \end{array} \right) = \begin{pmatrix} B_1 & O \\ O & B_2 \end{pmatrix}, \text{ 其中}$$

$$B_1 = \begin{pmatrix} a & 0 \\ 1 & a \end{pmatrix},$$

$$B_2 = \begin{pmatrix} b & 0 \\ 1 & b \end{pmatrix};$$

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} \mathbf{A}_1 & \\ & \mathbf{A}_2 \end{pmatrix} + \begin{pmatrix} \mathbf{B}_1 & \\ & \mathbf{B}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{A}_1 + \mathbf{B}_1 & \\ & \mathbf{A}_2 + \mathbf{B}_2 \end{pmatrix},$$

$$\mathbf{A}_1 + \mathbf{B}_1 = \begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix} + \begin{pmatrix} a & 0 \\ 1 & a \end{pmatrix} = \begin{pmatrix} 2a & 1 \\ 1 & 2a \end{pmatrix},$$

$$\mathbf{A}_2 + \mathbf{B}_2 = \begin{pmatrix} b & 1 \\ 1 & b \end{pmatrix} + \begin{pmatrix} b & 0 \\ 1 & b \end{pmatrix} = \begin{pmatrix} 2b & 1 \\ 2 & 2b \end{pmatrix},$$

$$\therefore \mathbf{A} + \mathbf{B} = \begin{pmatrix} 2a & 1 & 0 & 0 \\ 1 & 2a & 0 & 0 \\ 0 & 0 & 2b & 1 \\ 0 & 0 & 2 & 2b \end{pmatrix}.$$

$$\begin{aligned}
 \mathbf{ABA} &= \begin{pmatrix} \mathbf{A}_1 & \\ & \mathbf{A}_2 \end{pmatrix} \begin{pmatrix} \mathbf{B}_1 & \\ & \mathbf{B}_2 \end{pmatrix} \begin{pmatrix} \mathbf{A}_1 & \\ & \mathbf{A}_2 \end{pmatrix} \\
 &= \begin{pmatrix} \mathbf{A}_1 \mathbf{B}_1 \mathbf{A}_1 & \\ & \mathbf{A}_2 \mathbf{B}_2 \mathbf{A}_2 \end{pmatrix},
 \end{aligned}$$

$$\mathbf{A}_1 \mathbf{B}_1 \mathbf{A}_1 = \begin{pmatrix} a^3 + a & 2a^2 + 1 \\ a^2 & a^3 + a \end{pmatrix}, \quad \mathbf{A}_2 \mathbf{B}_2 \mathbf{A}_2 = \begin{pmatrix} b^3 + 2b & 2b^2 + 1 \\ 3b^2 & b^3 + 2b \end{pmatrix},$$

$$\therefore \mathbf{ABA} = \begin{pmatrix} a^3 + a & 2a^2 + 1 & 0 & 0 \\ a^2 & a^3 + a & 0 & 0 \\ 0 & 0 & b^3 + 2b & 2b^2 + 1 \\ 0 & 0 & 3b^2 & b^3 + 2b \end{pmatrix}.$$

例3 设 $A = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 2 & 1 \end{pmatrix}$, 求 A^{-1} .

解 $A = \left(\begin{array}{c|cc} 5 & 0 & 0 \\ \hline 0 & 3 & 1 \\ 0 & 2 & 1 \end{array} \right) = \begin{pmatrix} A_1 & O \\ O & A_2 \end{pmatrix},$

$$A_1 = (5), \quad A_1^{-1} = \left(\frac{1}{5} \right); \quad A_2 = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}, \quad A_2^{-1} = \begin{pmatrix} 1 & -1 \\ -2 & 3 \end{pmatrix};$$

$$\therefore A^{-1} = \begin{pmatrix} A_1^{-1} & O \\ O & A_2^{-1} \end{pmatrix} = \begin{pmatrix} 1/5 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -2 & 3 \end{pmatrix}.$$

例4 设分块矩阵 $A = \begin{pmatrix} B & O \\ C & D \end{pmatrix}$ ，其中

$B = B_{r \times r}, D = D_{s \times s}$ 均为可逆矩阵，求 A^{-1} .

解：由拉普拉斯定理知 $|A| = \begin{vmatrix} B & O \\ C & D \end{vmatrix} = |B| |D| \neq 0$

故 A 可逆. 设 A^{-1} 的分块形式为

$$A^{-1} = \begin{pmatrix} X & Y \\ Z & T \end{pmatrix}$$

其中， $X = X_{r \times r}, T = T_{s \times s}$

利用分块乘法有

$$\begin{aligned} \mathbf{A}\mathbf{A}^{-1} &= \begin{pmatrix} \mathbf{B} & \mathbf{O} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \begin{pmatrix} \mathbf{X} & \mathbf{Y} \\ \mathbf{Z} & \mathbf{T} \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{B}\mathbf{X} & \mathbf{B}\mathbf{Y} \\ \mathbf{C}\mathbf{X} + \mathbf{D}\mathbf{Z} & \mathbf{C}\mathbf{Y} + \mathbf{D}\mathbf{T} \end{pmatrix} = \mathbf{E} = \begin{pmatrix} \mathbf{E}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{E}_s \end{pmatrix} \end{aligned}$$

于是

$$\begin{cases} \mathbf{B}\mathbf{X} = \mathbf{E}_r \\ \mathbf{B}\mathbf{Y} = \mathbf{O} \\ \mathbf{C}\mathbf{X} + \mathbf{D}\mathbf{Z} = \mathbf{O} \\ \mathbf{C}\mathbf{Y} + \mathbf{D}\mathbf{T} = \mathbf{E}_s \end{cases} \Rightarrow \begin{cases} \mathbf{X} = \mathbf{B}^{-1} \\ \mathbf{Y} = \mathbf{O} \\ \mathbf{Z} = -\mathbf{D}^{-1}\mathbf{C}\mathbf{B}^{-1} \\ \mathbf{T} = \mathbf{D}^{-1} \end{cases}$$

故

$$A^{-1} = \begin{pmatrix} B^{-1} & O \\ -D^{-1}CB^{-1} & D^{-1} \end{pmatrix}$$

特别的, 当 $C=O$ 时, $A = \begin{pmatrix} B & O \\ O & D \end{pmatrix}$

$$A^{-1} = \begin{pmatrix} B^{-1} & O \\ O & D^{-1} \end{pmatrix}$$

这与对角形矩阵的结论是一致的.

§ 2.5.2 方阵的迹

定义 设 $A = (a_{ij})_{n \times n}$, 则 A 的迹为

性质
$$\text{tr}(A) = a_{11} + a_{22} + \dots + a_{nn}$$

- $\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$
- $\text{tr}(kA) = k \cdot \text{tr}(A)$
- $\text{tr}(AB) = \text{tr}(BA)$
- $\text{tr}(P^{-1}AP) = \text{tr}(A)$
- $A = \text{diag}(A_1, A_2, \dots, A_s)$, 则

$$\text{tr}(A) = \text{tr}(A_1) + \text{tr}(A_2) + \dots + \text{tr}(A_s)$$